STABILITY UNDER SPECIALIZATION IS EQUIVALENT TO BEING A UNION OF CLOSED SUBSETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The specialization relation and stability under specialization, together with the fact that closed subsets are stable under specialization. Set up in the author's section on specialization, immediately before the counted statement. Supplies one direction of the equivalence and the notion being characterized.

Lemma 0.1. *Let $T \subset X$ be a subset of a topological space X. The following are equivalent*

(1) *T is stable under specialization, and*
(2) *T is a (directed) union of closed subsets of X.*

Proof. Suppose that T is stable under specialization, then for all $y \in T$ we have $\overline{\{y\}} \subset T$. Thus $T = \bigcup_{y \in T} \overline{\{y\}}$ which is an union of closed subsets of X. Reciprocally, suppose that $T = \bigcup_{i \in I} F_i$ where F_i are closed subsets of X. If $y \in T$ then there exists $i \in I$ such that $y \in F_i$. As F_i is closed, we have $\overline{\{y\}} \subset F_i \subset T$, which proves that T is stable under specialization. □

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